ACCESS CONTROL

Presented by: Dr. **Brahim Ferik**

WHAT IS ACCESS CONTROL?

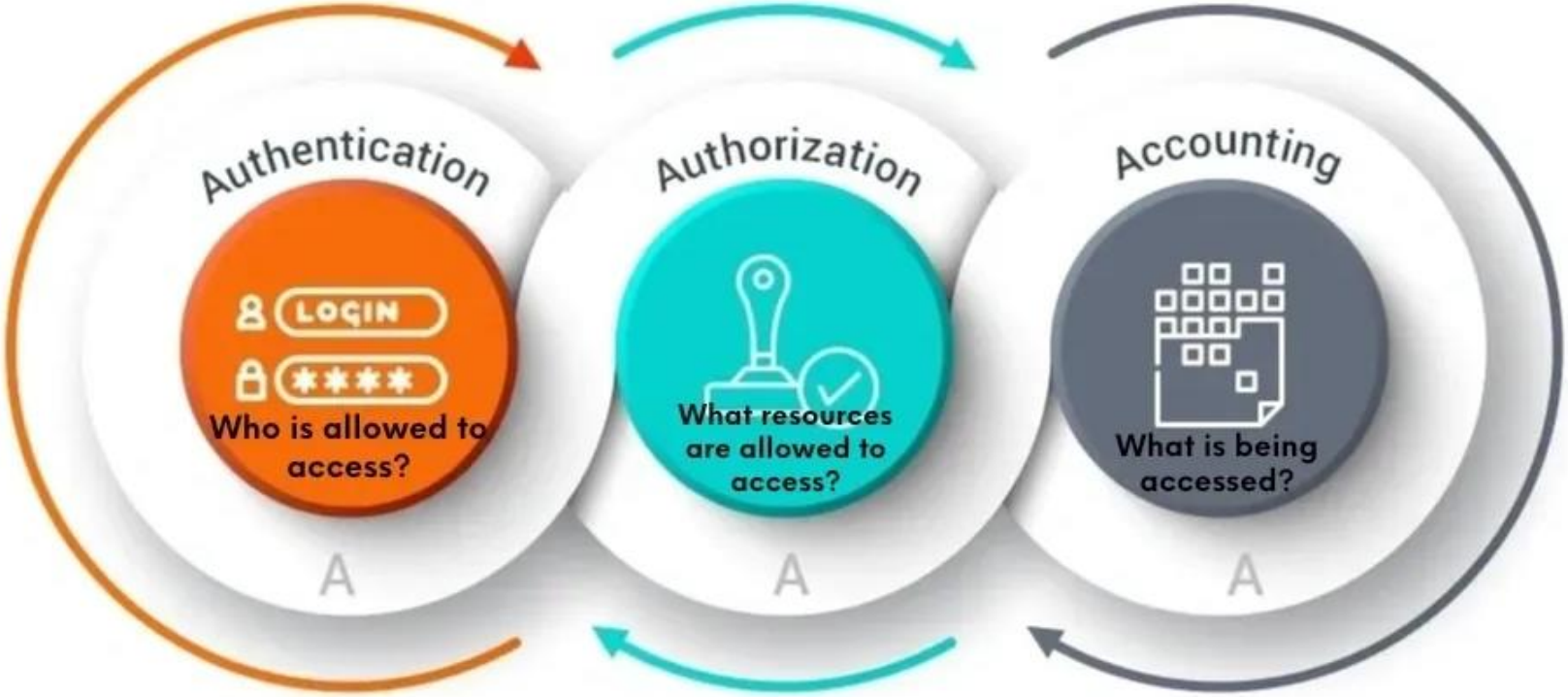
- **Access Control is the process of restricting and managing who can access resources and what actions they are allowed to perform.**

KEY GOALS

- ❑ **Prevent unauthorized access.**
- ❑ **Protect data, systems, and services.**
- ❑ **Enforce security policies.**

THE AAA SECURITY MODEL

- ❑ **Authentication**
- ❑ **Authorization**
- ❑ **Accounting**



AUTHENTICATION (AAA)

Common Authentication Methods :

- ❑ Passwords / Passphrases
- ❑ Smart cards
- ❑ Biometrics
- ❑ Multi-Factor Authentication (MFA)

AUTHORIZATION (AAA)

Authorization determines permissions after login :

- ❑ **User roles**
- ❑ **Group membership**
- ❑ **File and resource permissions**

ACCOUNTING (AAA)

Why Accounting Matters :

- ❑ **Detect attacks and misuse**
- ❑ **Support audits and compliance**
- ❑ **Investigate security incidents**

TYPES OF ACCESS CONTROL

- DAC – Discretionary Access Control
- RBAC – Role-Based Access Control
- MAC – Mandatory Access Control
- ABAC – Attribute-Based Access Control

ACCESS CONTROL: PASSWORD SECURITY ESSENTIALS

- **Passwords are a fundamental component of access control, serving as a primary authentication method. Strong passwords enhance security by preventing unauthorized access.**

LESSON OBJECTIVES

- ❑ Understand common vulnerabilities in passwords.
- ❑ Learn how attackers exploit weak passwords.
- ❑ Develop strategies to create and manage uncrackable passwords.
- ❑ Implement advanced authentication layers.

TOP MOST COMMON PASSWORDS 2025

Generations change, password habits remain

TOP MOST COMMON PASSWORDS 2025

TOP 10
MOST COMMON
PASSWORDS
IN 2023

****

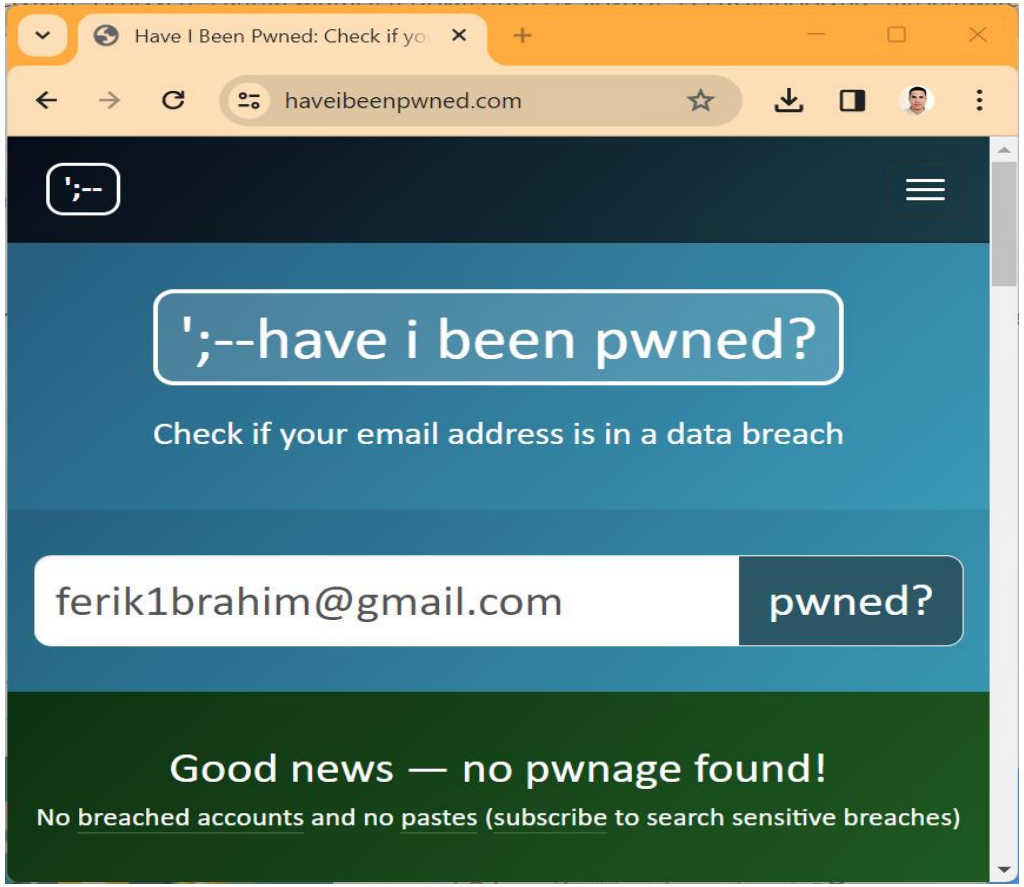
1	123456
2	123456789
3	qwerty
4	password
5	12345
6	qwerty123
7	1q2w3e
8	12345678
9	111111
10	1234567890

TOP MOST COMMON PASSWORDS 2025

- https://tinyurl.com/3ajp69ry
- https://tinyurl.com/4vv2mfcb
- https://tinyurl.com/2dxzxwpz

VERIFICATION OF PROTECTION?

haveibeenpwned.com



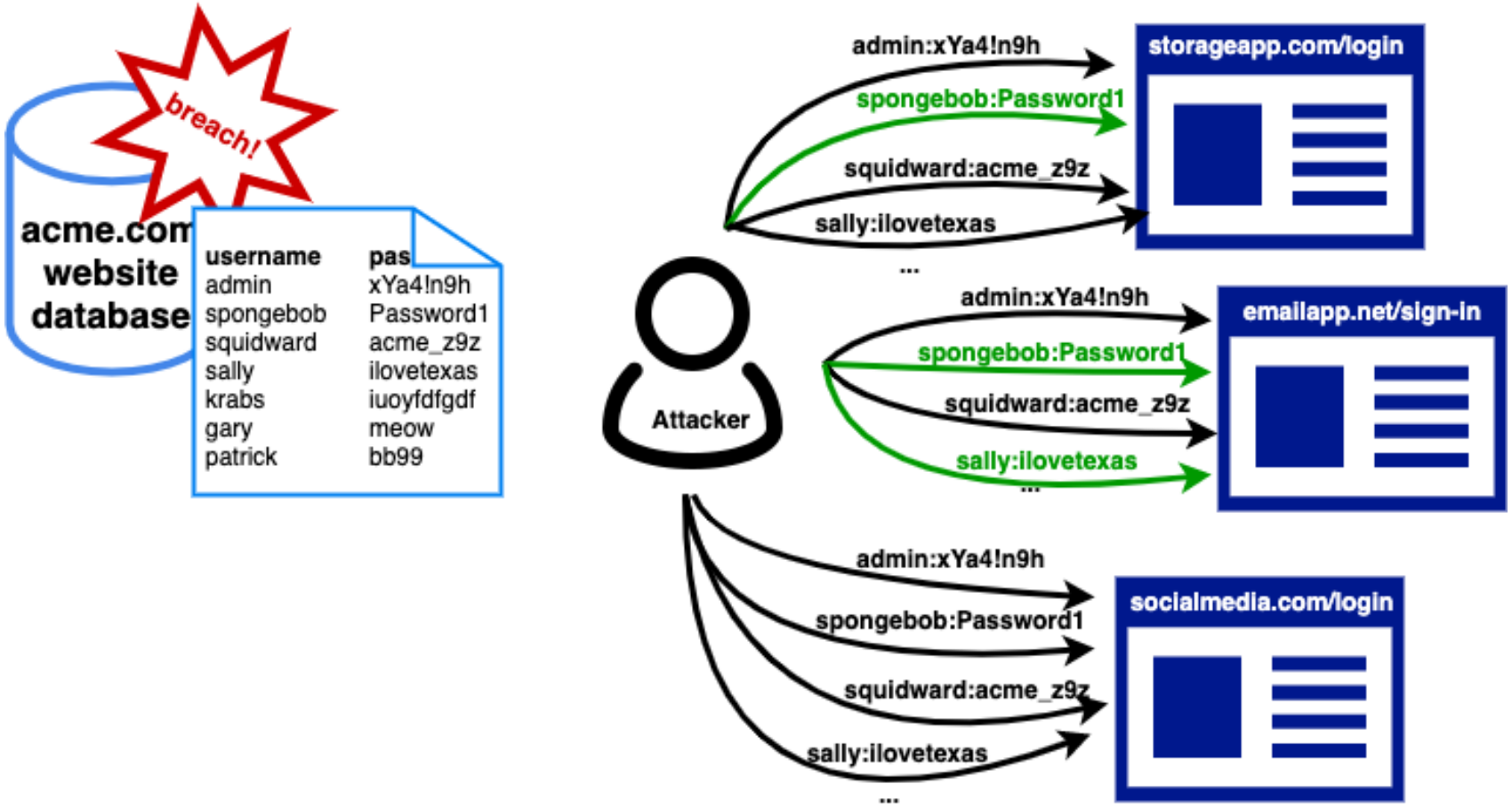
PASSWORD CRACKING SIMULATION

PASSWORD CRACKING METHODS

- **Key Techniques**

- ❑ **Dictionary Attacks.**
- ❑ **Combinator Attacks.**
- ❑ **Brute-force Attacks.**
- ❑ **Phishing/Social Engineering.**
- ❑ **Credential Stuffing.**

PASSWORD CRACKING METHODS



PASSWORD CRACKING METHODS

PASSWORD CRACKING METHODS

- https://www.openwall.com/john/

PASSWORD CRACKING METHODS

HASHCAT

- https://hashcat.net/hashcat/

PASSWORD CRACKING METHODS

- Applied Work (rar file)
- ❑ john.exe --list=formats
- ❑ Hash / Encryption Formats

PASSWORD CRACKING METHODS

- **Applied Work (rar file)**

- ❑ **john.exe --list=formats**
- ❑ **rar2john.exe protected.rar>hash.txt**
- ❑ **john.exe hash.txt**
- ❑ **john.exe --show hash.txt**

PASSWORD CRACKING METHODS

- **Applied Work (rar file)**

- rar2john.exe protected.rar
- hashcat.exe -m 13000 -a3 value_hash ?d?d?d?d
- hashcat.exe –show -m 13000 value_hash
- https://tinyurl.com/3ca4kps6

PASSWORD CRACKING METHODS

- **Applied Work (Zip file)**

- ❑ **john.exe –list=formats**
- ❑ **zip2john.exe protected.zip>zip.hash**
- ❑ **john.exe zip.hash**

PASSWORD CRACKING METHODS

PASSWORD CRACKING METHODS

- **Applied Work (PDF, DOCX, XLSX, ...)**
- ❑ **Practical Exercises.**
- ❑ **Prepare a wordlist containing common passwords.**

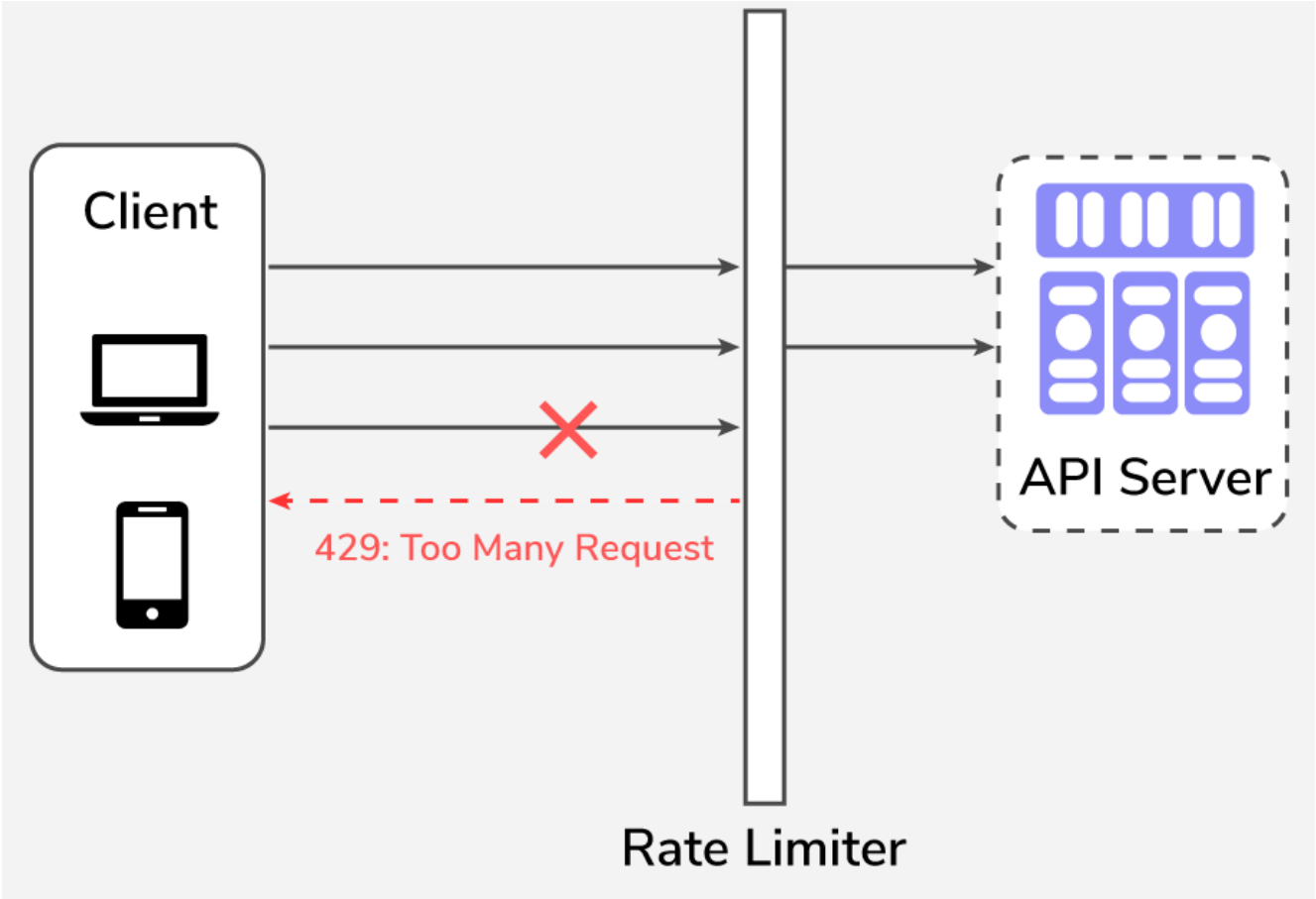
HOW TO PREVENT BRUTE FORCE PASSWORD HACKING?

- Never use information that can be found online (like names of family members).
- Have as many characters as possible.
- Combine letters, numbers, and symbols.
- Be different for each user account.
- Do not use common patterns.

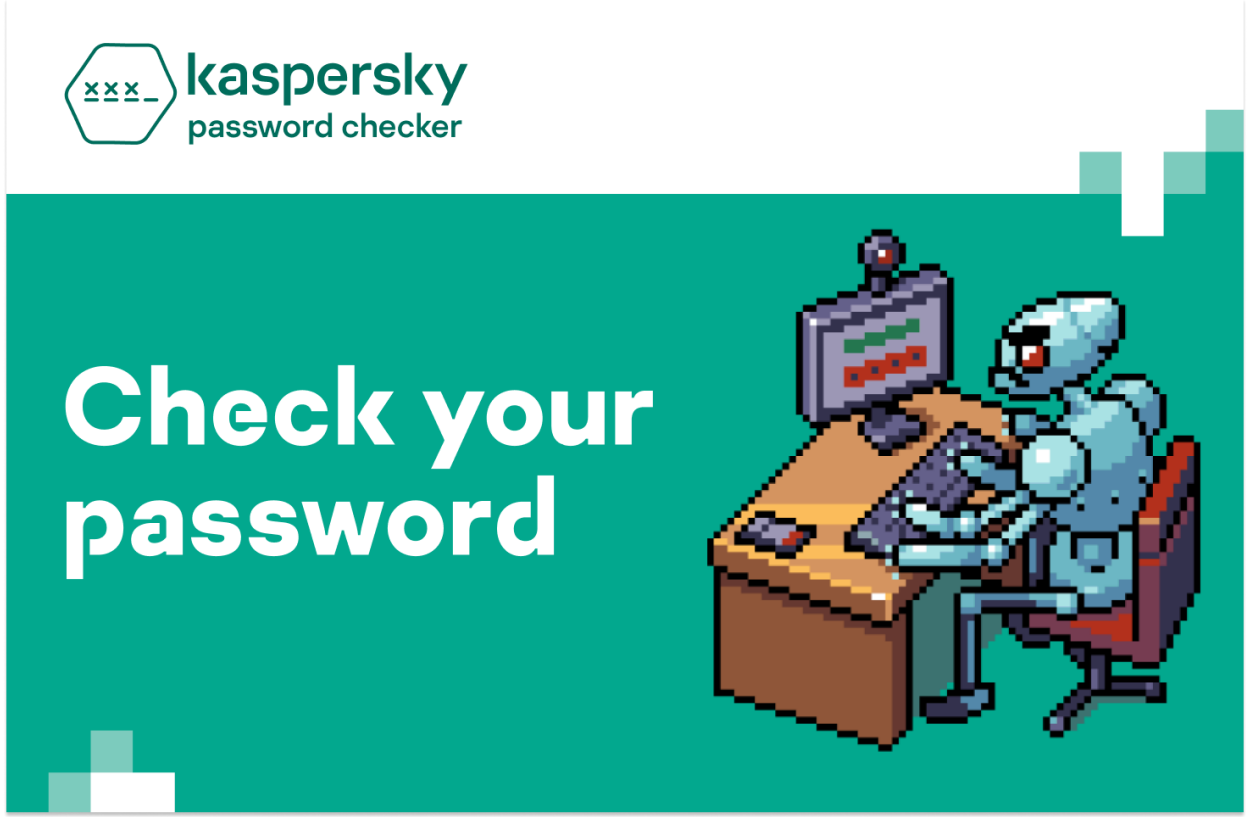
HOW TO PREVENT BRUTE FORCE PASSWORD HACKING?

- ❑ **Lockout policy**
- ❑ **Progressive delays**
- ❑ **Captcha**
- ❑ **Requiring strong passwords**
- ❑ **Two-factor authentication**

HOW TO PREVENT BRUTE FORCE PASSWORD HACKING?

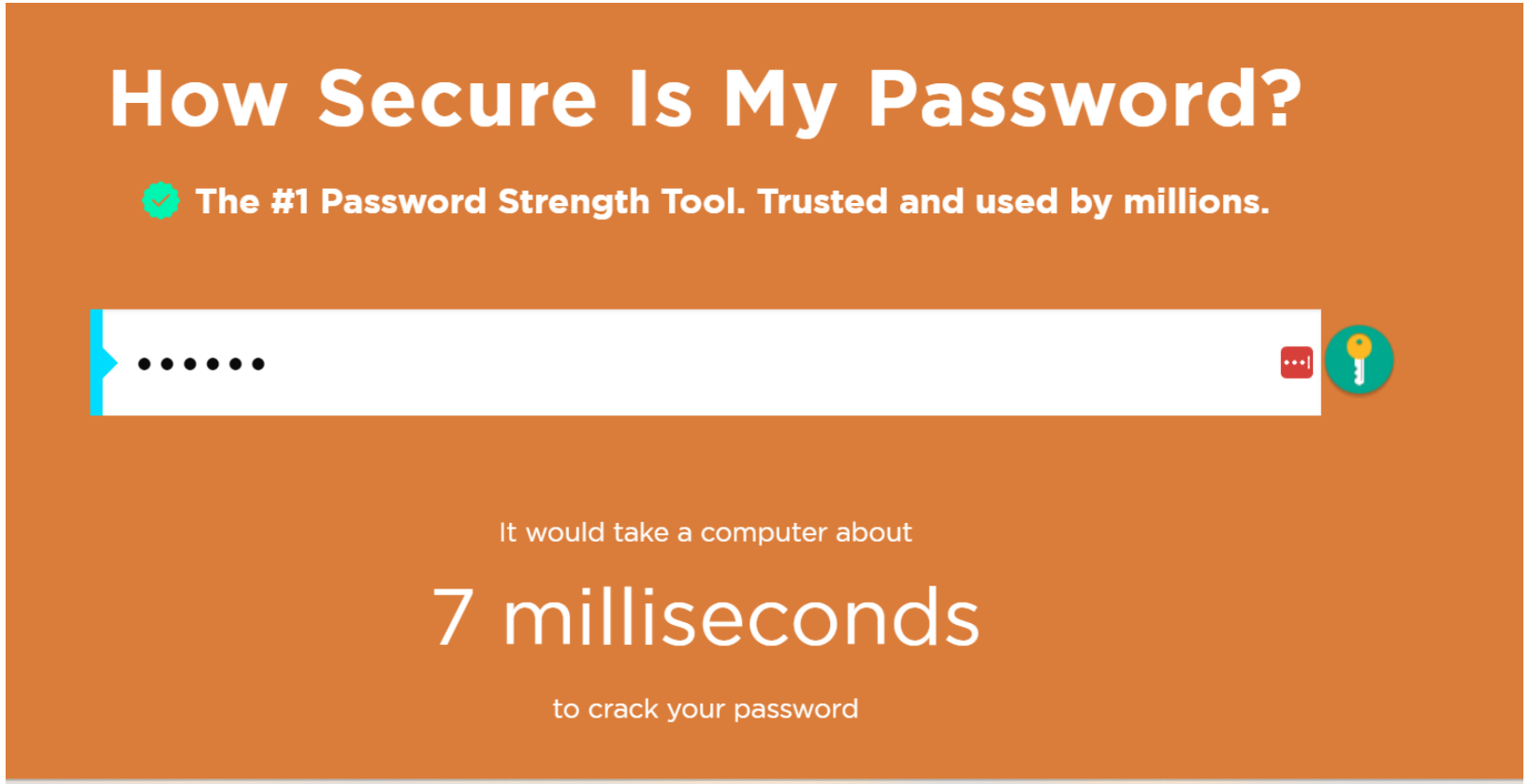


CHECK YOUR PASSWORD



password.kaspersky.com

HOW SECURE IS MY PASSWORD?



www.passwordmonster.com howsecureismypassword.net

PASSWORD GENERATION STRATEGIES

- ❑ **Use of Password Managers**
- ❑ **Passphrases**
- ❑ **Random Character Combinations**
- ❑ **Keyboard Patterns**

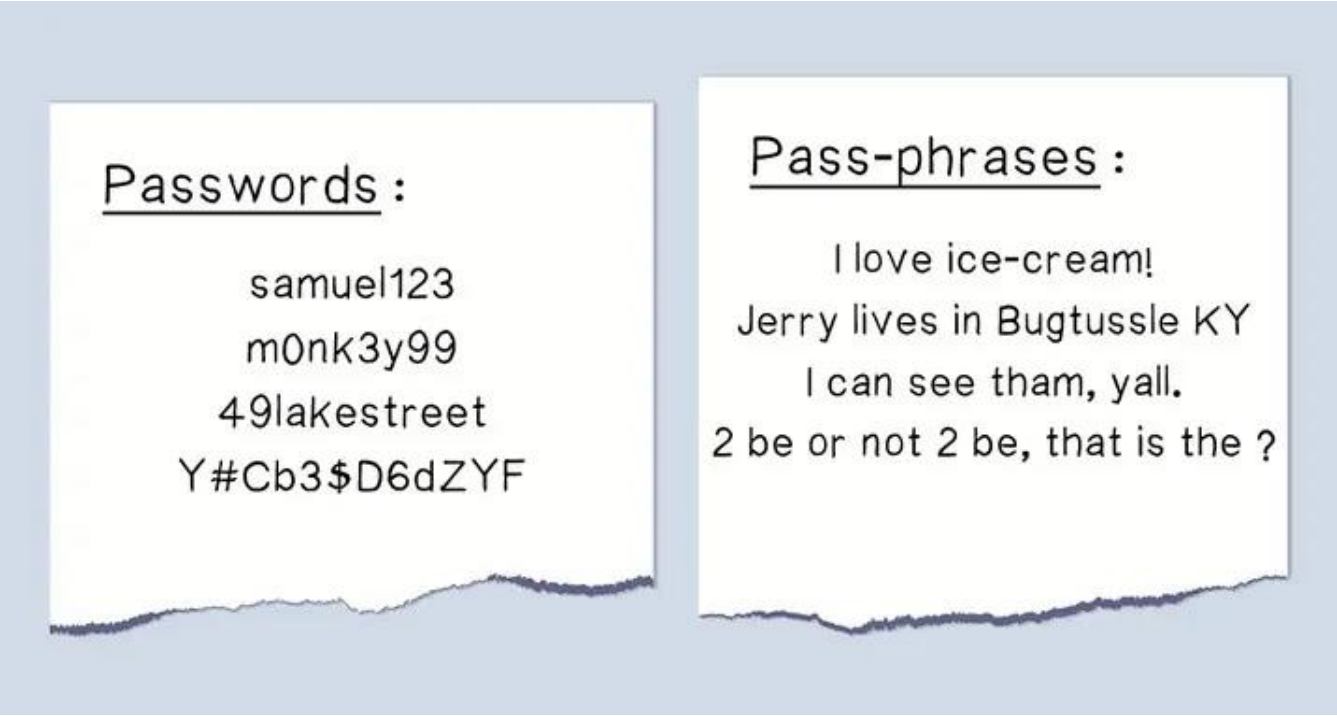
PASSWORD GENERATION STRATEGIES

- **Example of a Strong Password**

G7#pX9@wZ2!qL3_s

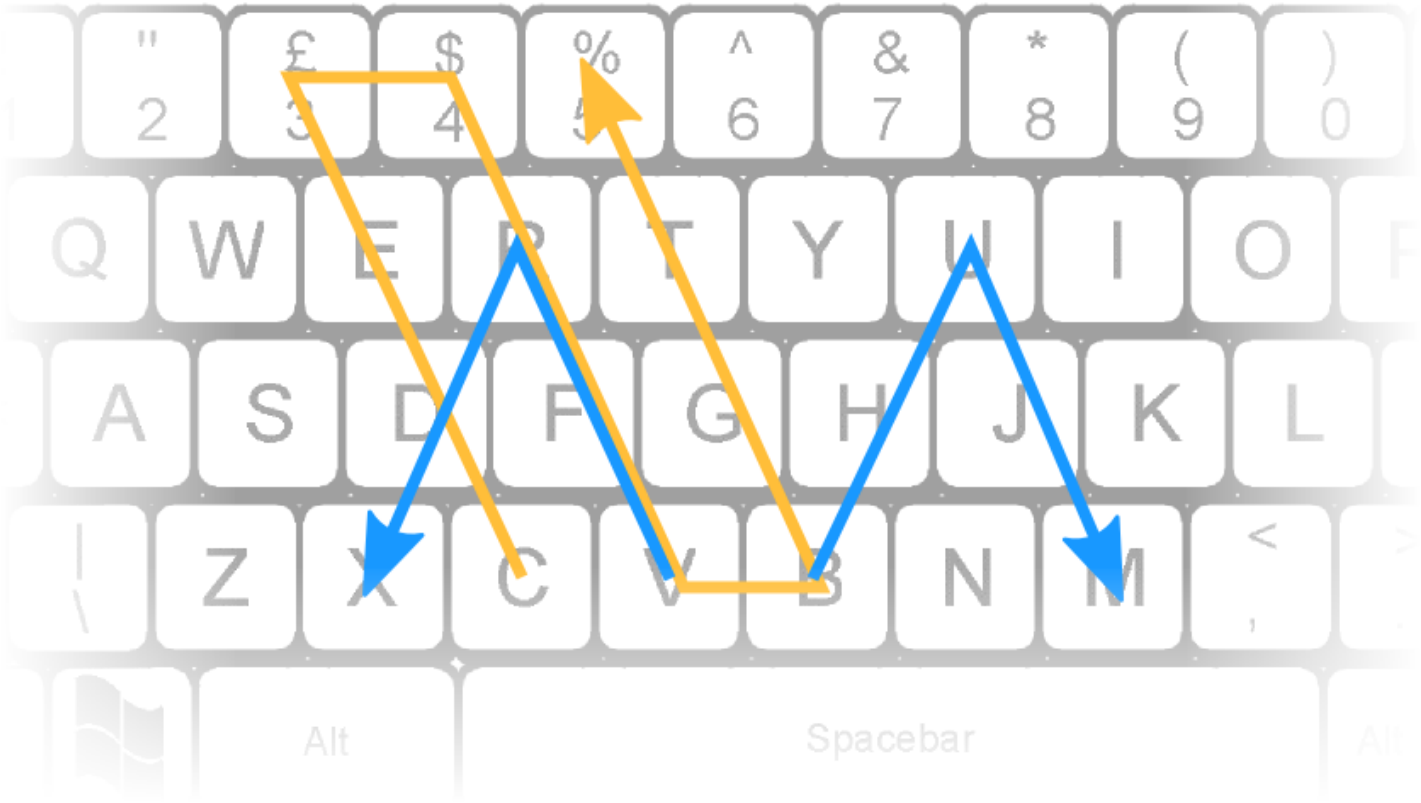
PASSWORD GENERATION STRATEGIES

- **Passphrases**



PASSWORD GENERATION STRATEGIES

- **Keyboard Patterns**



PASSWORD POLICIES FOR LOCAL WINDOWS ACCOUNTS - DEMO

- ❑ **Password Policy**
- ❑ **Account Lockout Policy**

WINDOWS USER MANAGEMENT

- **The Syntax of the Command**

net user ?

- **View All Users on the System**

net user

WINDOWS USER MANAGEMENT

- Adding a User without a Password

  net user username /add

- Set Password

  net user username password

- Set Password with Confirm

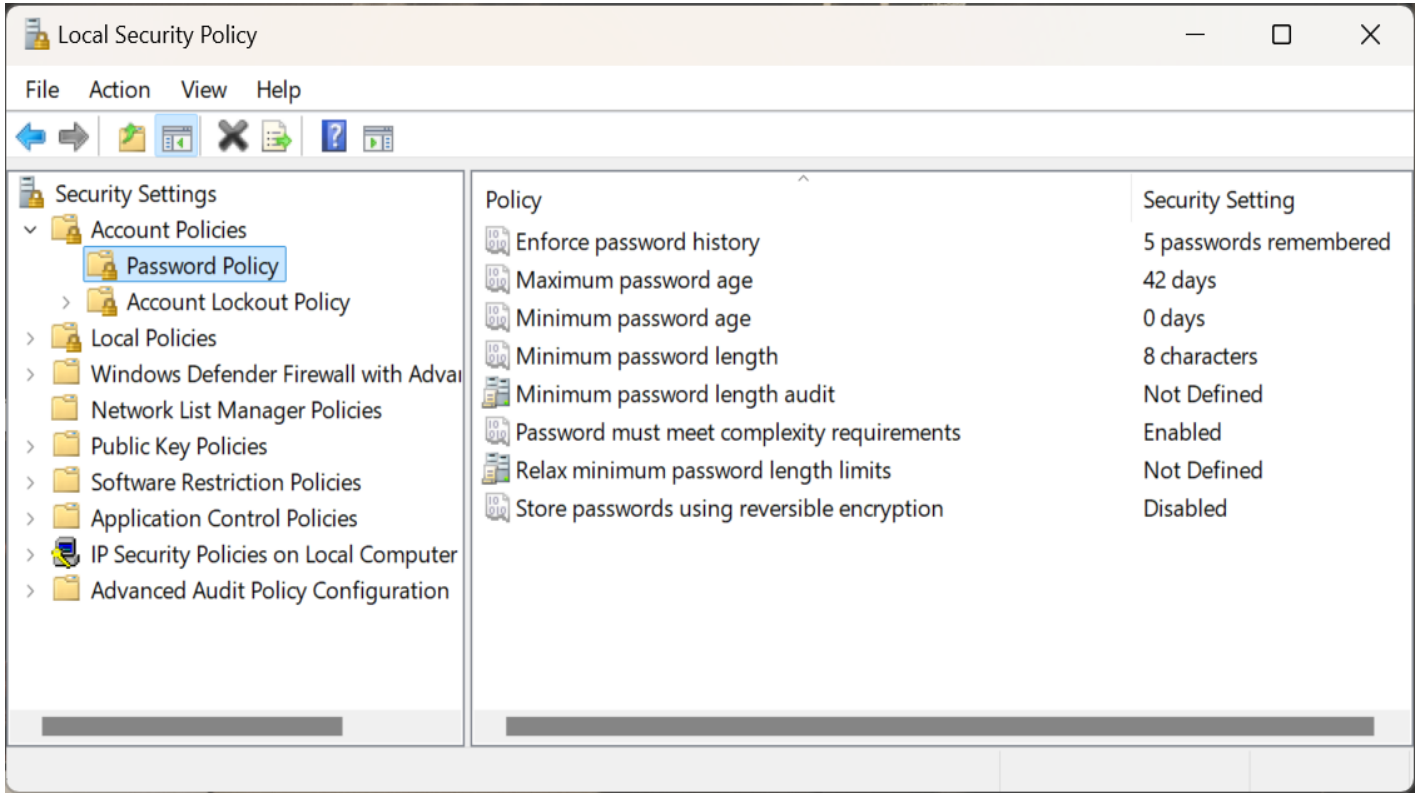
  net user username *

- Deleting a Specific User

  net user username /delete

PASSWORD POLICIES

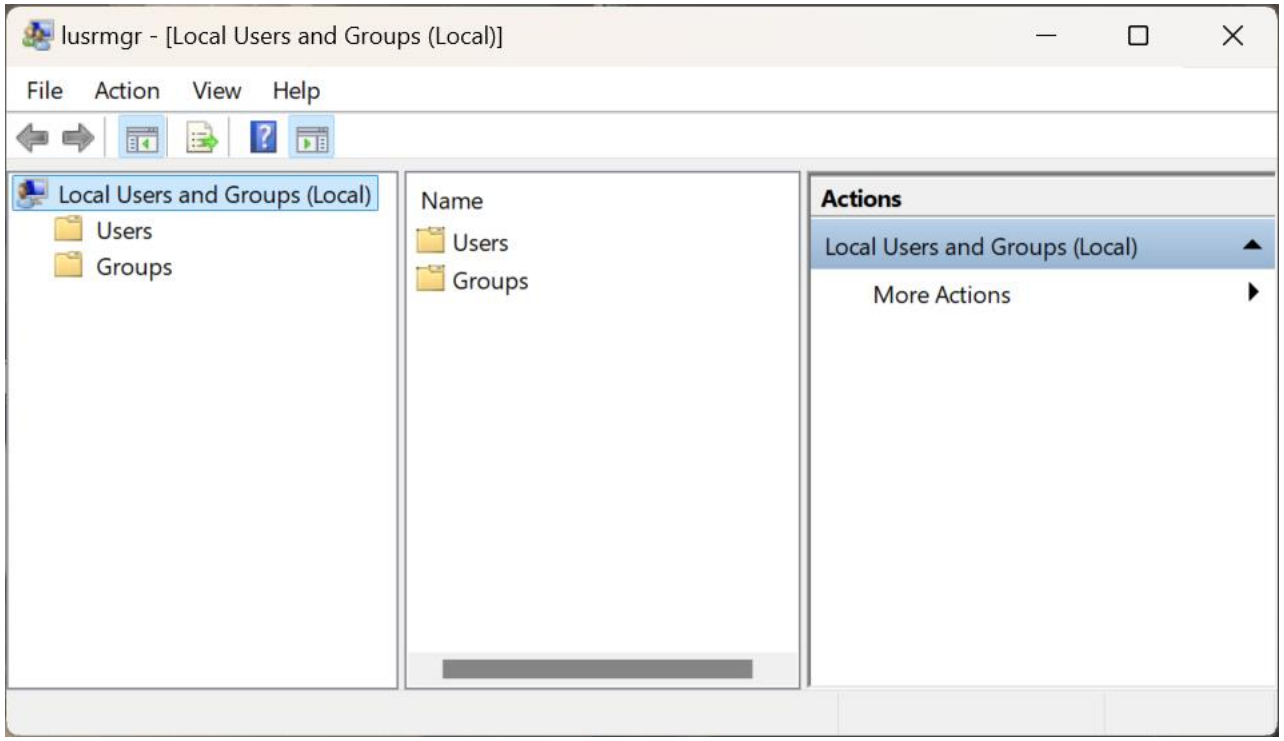
- ❑ **Search Local Security Policy**
- ❑ **Win + R : secpol.msc**
- ❑ **secpol.msc CMD**



PASSWORD POLICIES FOR LOCAL WINDOWS ACCOUNTS

- ❑ **Enforce password history (3 passwords)**
- ❑ **Maximum password age (90 days)**
- ❑ **Minimum password length (8)**
- ❑ **Password must meet complexity requirements (Enable)**

- Computer Management > System Tools > Local Users and Groups.
- Win + R : lusrmgr.msc
- lusrmgr.msc CMD



ACCOUNT LOCKOUT POLICY

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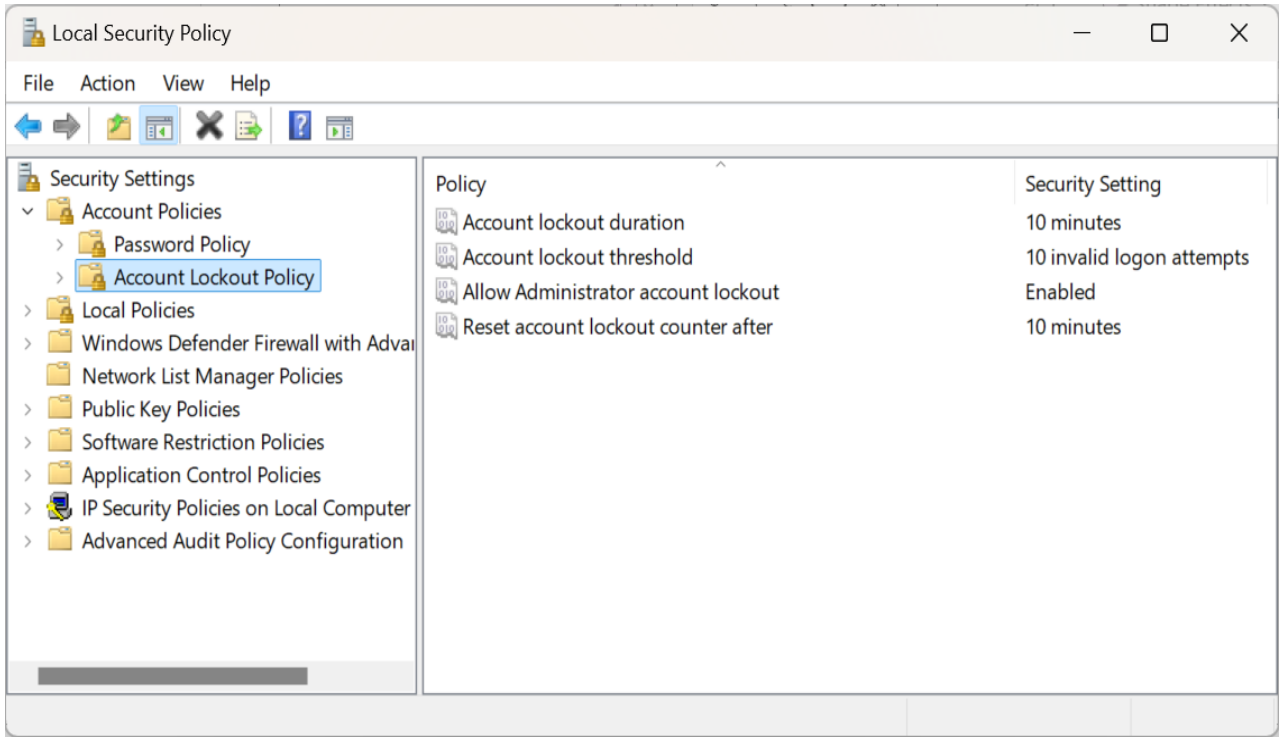

ACCOUNT LOCKOUT POLICY

- ❑ **Account lockout threshold: 3 invalid logon attempts.**
- ❑ **Account lockout duration: 15 minutes.**
- ❑ **Reset account lockout counter after: 5 minutes.**

ENABLE TWO-FACTOR AUTHENTICATION (2FA)

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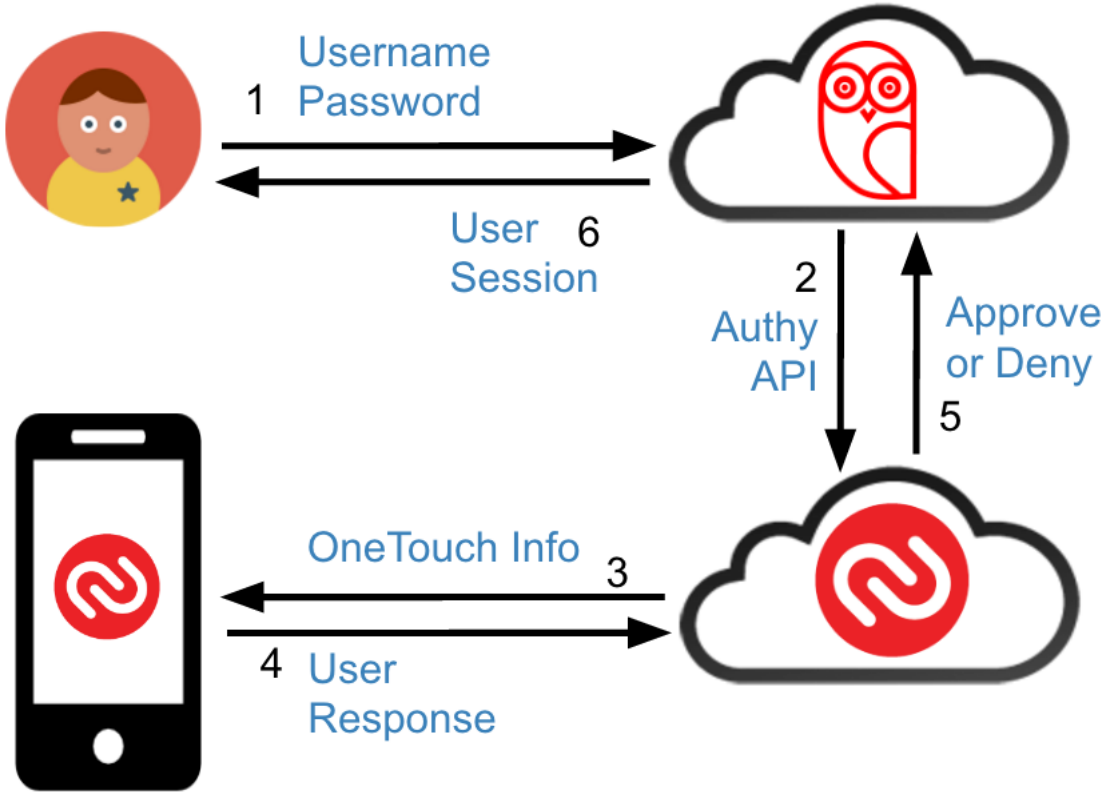
Google Authenticator

Microsoft Authenticator

ENABLE TWO-FACTOR AUTHENTICATION (2FA)

AUTHY

ENABLE TWO-FACTOR AUTHENTICATION (2FA)



PASSWORD MANAGEMENT TOOLS

- **Local vs. Cloud Managers**

- ❑ **Local (Offline): KeePass, Bitwarden**

- ❑ **Cloud-Based: 1Password, LastPass**

Keepass

bitwarden

USE A PASSWORD MANAGER

LastPass

Keepass

APPLIED PROGRAMMING TASKS: EXPLORING PASSWORD SECURITY ESSENTIALS

A. Develop a Password Strength Verification Tool.

B. Build a Secure Password Generator.

C. Implement Two-Factor Authentication.

D. Analyze Breached Passwords.

E. Passphrase Password Generator.

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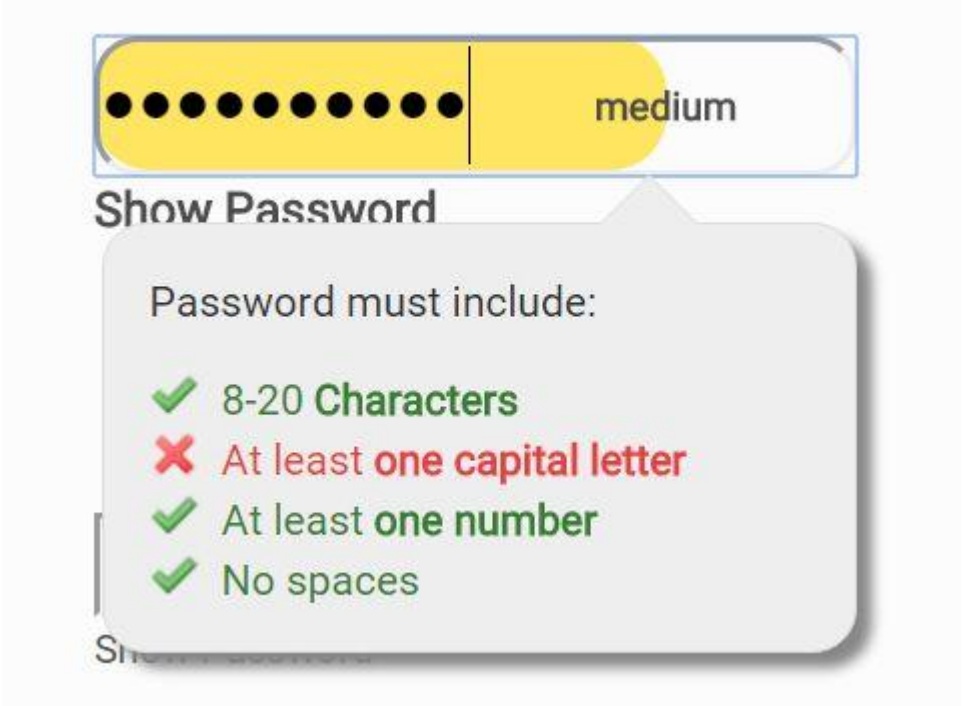
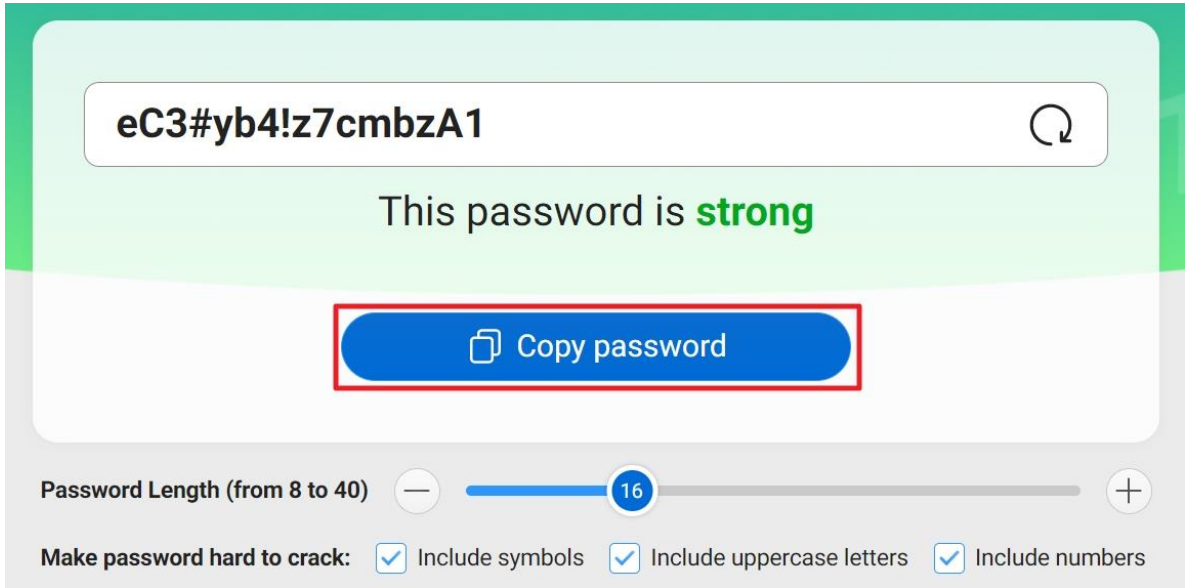


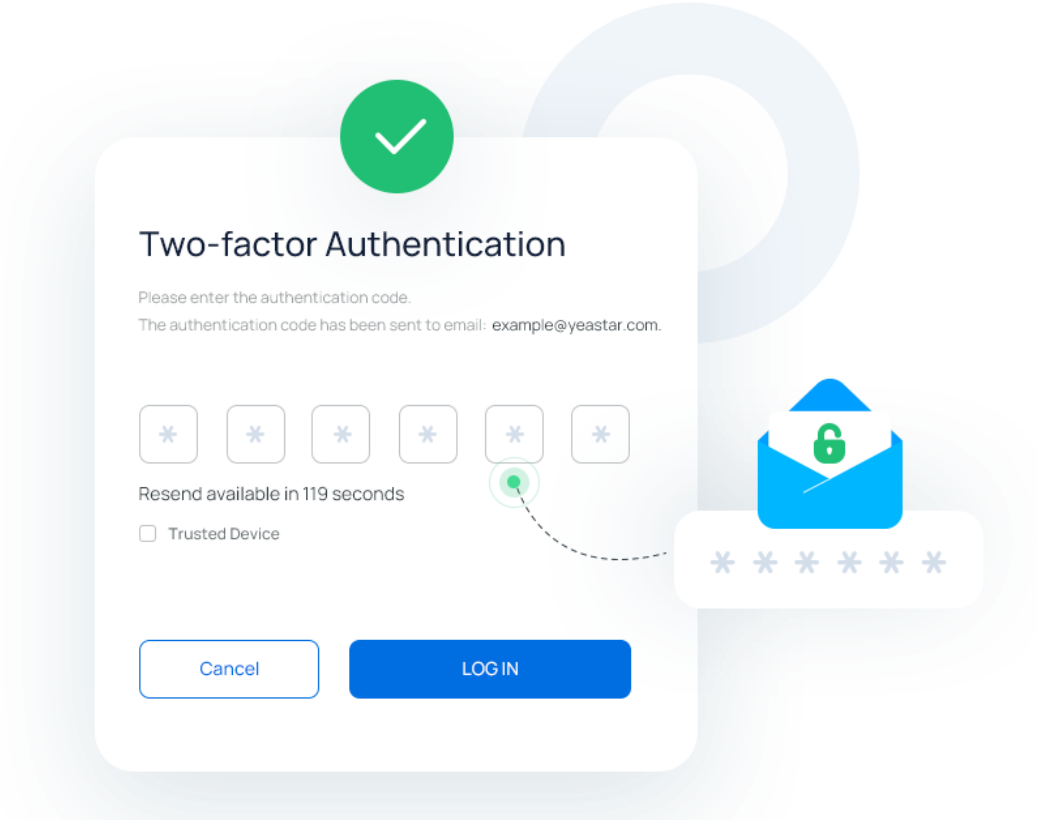
APPLIED PROGRAMMING TASKS: EXPLORING PASSWORD SECURITY ESSENTIALS

B. Build a Secure Password Generator.



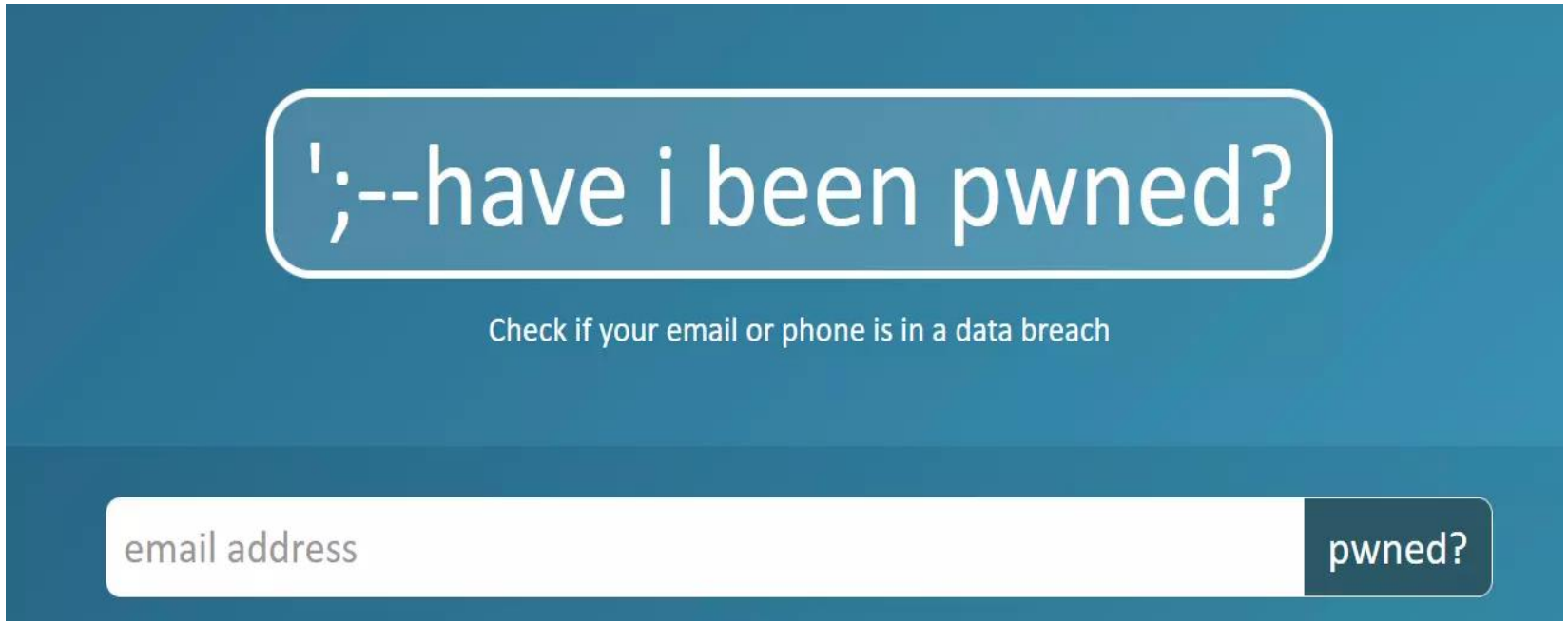
APPLIED PROGRAMMING TASKS: EXPLORING PASSWORD SECURITY ESSENTIALS

C. Implement Two-Factor Authentication.



APPLIED PROGRAMMING TASKS: EXPLORING PASSWORD SECURITY ESSENTIALS

D. Analyze Breached Passwords.



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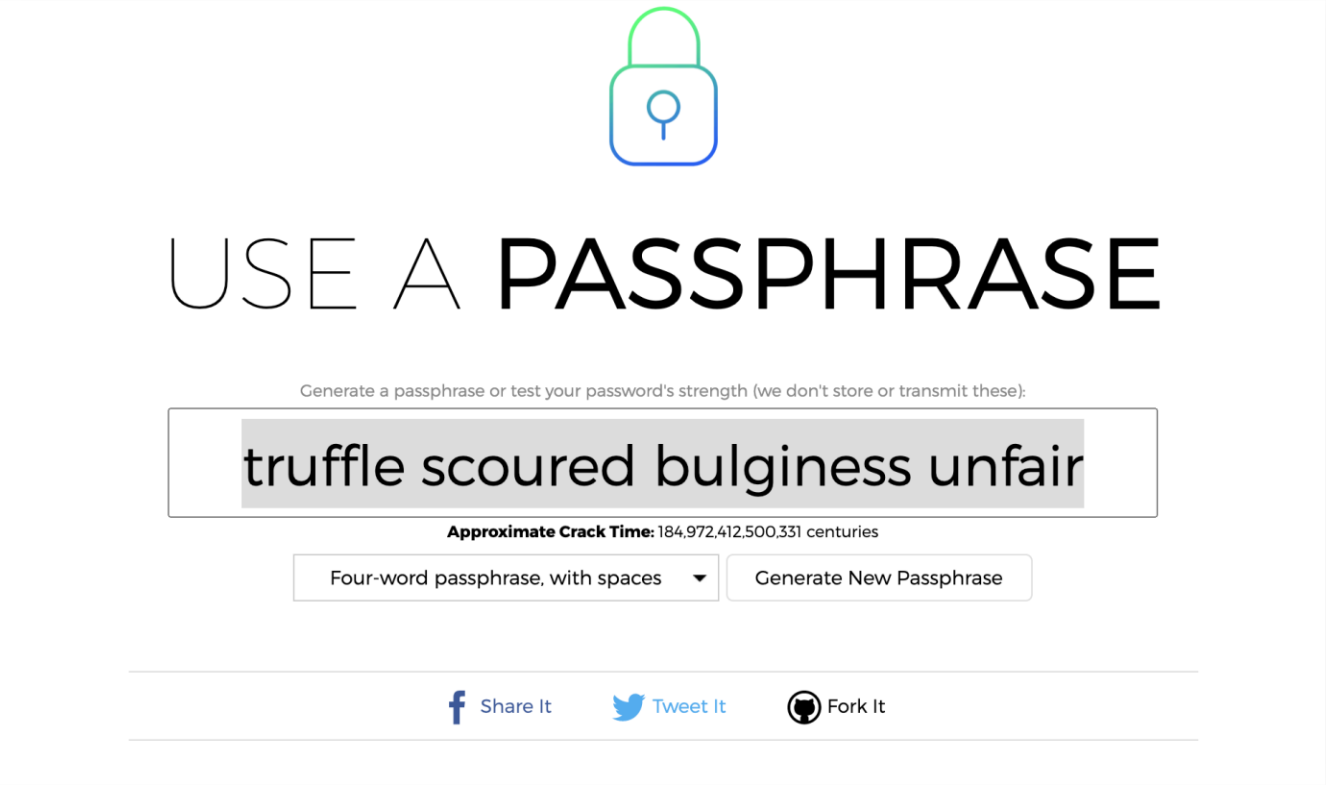
E. Passphrase Password Generator.